

Jp Varanasi

Phone: +44 7448 807707 | Email: jayaprakashvaranasi@gmail.com | Website: jpvaranasi.com | GitHub: github.com/jpvaranasi | LinkedIn: linkedin.com/in/jpvaranasi

Personal Statement

BSc Computer Science student combining software development with hands-on engineering leadership, from Python and Django web applications to React.js/TypeScript, AI/ML and C++/C# development. Built real-time collaborative applications and predictive AI models, and led a 30-person Formula Student team as Team Principal alongside a Digital Mentor role.

Skills & Certifications

Languages & Frameworks: Python | C++ | C# | TypeScript | React.js/Next.js | Django | HTML/CSS | Vite | Tailwind CSS

Cloud, AI/ML & Data: AWS Solutions Architect – Associate | Firebase | MongoDB | Machine Learning

Data Analysis: SQL | MoTeC

Tools: Git | Liveblocks | Clerk | ConvexDB | Recharts

Experience

Team Principal | USW Racing

Jan 2026 - Present

- Leading a 6 Department Formula Student Team across engineering, software, and operations, setting technical direction and overseeing budget/sponsorship strategy while acting as a liaison with leads from F1 Teams. Managing a team of 30 members across 6 team leads and 6 deputies, with a budget of over £50k and 13 sponsorships.
- Overseeing project planning, technical decision making and delivery while managing collaboration between engineering teams.
- Managing competing technical priorities, troubleshooting issues and making decisions under extreme time and resource constraints.
- Representing the team externally to sponsors, university leadership, and Formula Student UK Judges.

Software Lead | USW Racing

Sep 2025 - Present

- Led the software development team responsible for designing and delivering bespoke software for a FSUK race car.
- Designed and developed software solutions in collaboration with engineers to translate vehicle requirements into practical technical implementations.
- Coordinated software development, task allocation and technical priorities across a team of 4, ensuring 8+ projects were delivered to agreed requirements and deadlines while collaborating with 3 other engineering departments.
- Debugged and resolved technical issues across software and hardware interfaces in a time and resource constrained engineering environment.
- Actively pursuing Module Integration for Computer Science where FSUK Software Department would act as a module option for second year students worth 20 credits.

Developer | USW Racing

Jan 2025 - Present

- Designed and implemented fuel efficiency simulations to evaluate consumption across varying RPM and load conditions, improving engine performance analysis.
- Built and maintained live telemetry pipelines using data from 25+ sensors for real-time monitoring of vehicle data, enabling the race team to make rapid, data-driven decisions during testing and competition.
- Configured and optimized the ECU (Engine Control Unit) for performance, reliability, and compliance with Formula Student regulations.
- Collaborated with mechanical and electrical sub teams to integrate software solutions with sensor systems, ensuring accurate data acquisition and logging.
- Developed Python based tools to process and visualize sensor/CSV data for pre-race analysis; ran 10+ datasets with MoTeC for telemetry and data analysis.
- Built bespoke software for the team such as Attendance Monitoring System (AMS), custom website scraper, custom attendance system using one time codes (data handling via Firebase) and an official responsive team website as a solo developer. The scraper saved 8+ hours per monthly analysis; the AMS supported 30+ people every week for a year.

Digital Mentor | USWSU

Sep 2025 - Present

- Supported 400+ students in navigating core digital platforms such as Blackboard, timetabling systems, and online learning tools, ensuring smooth access to academic resources.
- Provided desk-based technical guidance, troubleshooting issues with course materials, schedules, and submissions.
- Delivered one-to-one mentoring through 4 mentoring sessions per week to help students build digital confidence, improve study workflows, and adapt to university systems.
- Collaborated with academic staff to streamline support processes and enhance the student digital experience.

Workshop Lead - Hanoi, Vietnam | USW

Jun 2025 - Jul 2025

- Showcased technology developed to detect flood probability levels in UK and relayed data on impact on Vietnamese Rivers/Floods (Mekong Delta).
- Delivered coding/effective thinking workshops to 50+ students in an international setting overcoming language barriers and cultural differences.

Projects

Wordbuds – React.js, Liveblocks, Clerk, ConvexDB

- Developed a real-time collaborative document editor using React.js, Liveblocks, Clerk and ConvexDB, focusing on privacy, component-based UI design and modern frontend architecture.
- Implemented the application as part of a 4 person development team, taking responsibility for coordinating the team, allocating tasks and ensuring concurrently assigned work was completed effectively.
- Contributed across the development lifecycle, coordinating frontend development and collaborative functionality while maintaining consistency between independently developed components.
- Worked within a shared real-time application environment, requiring careful coordination of features and development tasks to maintain a cohesive user experience.

F1 Suite (in development) – F1 SaaS | AI/ML, React.js

- Developing a comprehensive F1 SaaS platform using React.js, combining motorsport data analysis with AI/ML driven tools for race strategy and vehicle performance analysis.
- Developing a machine-learning tyre degradation model using multi-parameter input, testing against numerous datasets to analyse degradation behaviour and forecast tyre performance across a stint.
- Using multiple data inputs to investigate tyre degradation trends and support analysis of optimal stint strategies, with the aim of turning raw telemetry into actionable race information.
- Building a personalised F1 Race Engineer capable of analysing live races and custom telemetry data to provide recommendations around setup, pit timing and driver instructions.

DigiFob – SaaS Web App | Django, Python, HTML/CSS

- Built a digital key and access management SaaS application using Django and Python, providing an alternative approach to managing physical access credentials.
- Designed a role-based system supporting 4 distinct user roles: parent, student, staff and admin, with functionality tailored around different access requirements.
- Implemented session control functionality within the Django backend to manage authenticated users and control access to application functionality.
- Developed the frontend using HTML/CSS, creating a responsive interface designed to work across different screen sizes.

Jarvis – Voice-Controlled Desktop Assistant | Python

- Developed a Python based voice-controlled desktop assistant capable of interpreting voice commands and interacting with system functionality.
- Implemented voice recognition alongside automated system controls, allowing the assistant to execute tasks based on spoken input rather than traditional desktop interaction.
- Integrated practical utilities including weather updates, web search, reminders and task automation, bringing multiple functions together within a single assistant.
- Designed the project around extensibility, allowing additional commands and automated functionality to be incorporated into the assistant as required.

Education

2024-27 | BSc Computer Science (Hons) | University of South Wales | Predicted: First Class Honours

2022-24 | A Levels | St Cyres Comprehensive School

Leadership & Involvement

Coding Society President | Founder Google Developer on Campus Society | Elected Course Representative | Elected Student Voice Representative | PASS Leader
STEM Ambassador | Course Structure Advisor | USW Representative - Cardiff Innovation